

Calculus Language Quiz

5 questions. Choose the best answer for each.

Question 1

Which of the following is equivalent to $\log_2 16$?

- a) $\log_{10} \frac{16}{\log_{10} 2}$
- b) $2 \log_2 8$
- c) $\log_2 8 + \log_2 2$
- d) All of the above

Question 2

If $f(x) = x^2 - 3$ and $g(x) = 2x + 1$, what is $(f \circ g)(x)$?

- a) $2x^2 - 5$
- b) $4x^2 + 4x - 2$
- c) $(2x + 1)^2 - 3$
- d) $2(x^2 - 3) + 1$

Question 3

The graph of $y = f(x - 2) + 3$ is obtained from $y = f(x)$ by:

- a) Shifting left 2 units and up 3 units
- b) Shifting right 2 units and up 3 units
- c) Shifting left 2 units and down 3 units
- d) Shifting right 2 units and down 3 units

Question 4

Using the half-angle formula, $\sin^2\left(\frac{\pi}{8}\right)$ equals:

- a) $\frac{1 - \frac{\sqrt{2}}{2}}{2}$
- b) $\frac{1 - \frac{\sqrt{2}}{2}}{4}$
- c) $\frac{1 + \frac{\sqrt{2}}{2}}{2}$
- d) $\sqrt{\frac{1 - \frac{\sqrt{2}}{2}}{2}}$

Question 5

Which function has a domain of all real numbers except $x = 0$?

- a) $f(x) = \ln x$
- b) $f(x) = \frac{1}{x}$
- c) $f(x) = \sqrt{x}$
- d) $f(x) = e^x$

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